JAIN DEEMED-TO-BE UNIVERSITY | FACULTY OF ENGINEERING AND TECHNOLOGY

TEACHING LEARNING EVALUATION PLAN (TLEP)

Academic Year 2026–2027

25ESHS01 – DESIGN THINKING

CSE–AIML

Particulars	Details
Name of Department / School	Computer Science & Engineering
Programme	CSE – Artificial Intelligence and Machine Learning (CSE-AIML)
Year / Semester	II Year (as per programme curriculum; semester to be confirmed by department)
Course Code and Title	25ESHS01 – Design Thinking
Credit Value	2
Total Notional / Contact Hours Available	60 hours
Contact Hours per Week	4 hours
LTPE Distribution	0–0–4–0
Assessment Model	100% Continuous Assessment
Number of Assessment Components	Unit-wise Exercise Portfolio (80%) + Comprehensive Design Challenge Project (20%)
Faculty In-charge	Dr. Sujeet S. Jagtap

Course design principle: Phase-1 and Phase-2 establish the common, department-agnostic design-thinking foundation. Phase-3 and Phase-4 specialise the process for CSE-AIML: each student develops a domain-specific, dataset-grounded Small Language Model (SLM) prototype and a documented pathway for advancing it in a later Professional, Entrepreneurship, or Research track.

1. Course Description and Intent

Design Thinking introduces human-centred problem discovery, empathy, user research, ideation, prototyping, validation, feasibility, and entrepreneurship. In CSE-AIML, the course retains the general foundations in the early phase and then applies the same process to a domain-specific language-model engineering challenge. Students start with a human need and a dataset, not a model architecture. They identify an underserved user or workflow, formulate a point-of-view (POV) and How Might We (HMW) statement, select and inspect a relevant dataset, design a small domain-specific language model, create and test a prototype, and communicate its value and limitations.

By the end of the semester, every student is expected to have an individual SLM artefact or individually attributable SLM implementation within a team workflow. The artefact may be a small model trained from scratch when feasible, or a compact domain-adapted model trained/fine-tuned on the selected dataset. The student must document the dataset, data governance, model approach, evaluation evidence, limitations, and next-step plan. The course does not promise production-grade deployment; it establishes a validated prototype and an evidence-based design brief that can be extended in later semesters.

2. Course Learning Objectives

- Apply human-centred design principles to identify and frame real-world problems before selecting a technical solution.
- Demonstrate empathy for diverse users, particularly underserved communities, through observation, interviews, persona development, and qualitative/quantitative research.
- Generate and select design concepts using structured ideation methods while considering social, cultural, ethical, economic, and technical constraints.
- Prototype and validate a domain-specific SLM concept through dataset preparation, model selection/adaptation, evaluation, and feedback-driven iteration.
- Work collaboratively while maintaining individual ownership, reflective documentation, responsible AI practice, and clear technical communication.
- Connect the SLM prototype to a plausible continuation pathway in the Professional, Entrepreneurship, or Research track.

3. Course Outcomes (COs)

CO	Course Outcome	Indicative Bloom's Level
CO1	Identify the principles and stages of design thinking through experiential exercises and explain why human-centred problem discovery precedes solution selection.	Understand
CO2	Conduct user observation, empathy mapping, stakeholder analysis, interviews and basic qualitative/quantitative research to formulate a supported POV/HMW statement.	Apply/Analyse
CO3	Generate, compare and justify domain-specific solution concepts, including dataset and model-design choices, using structured ideation and evidence.	Analyse/Create
CO4	Construct a proof-of-concept domain-specific SLM using an appropriate dataset and model adaptation/training workflow, documenting design decisions.	Apply/Create
CO5	Evaluate the prototype with a documented test plan, suitable task/model metrics, user or peer feedback, and feasibility/limitations analysis.	Evaluate
CO6	Communicate the SLM's value proposition, responsible-use boundaries and next-step plan for Professional, Entrepreneurship or Research-track development.	Create/Evaluate

4. Programme Outcomes (POs), PSOs and Knowledge Profile

The course contributes most directly to PO2 (Problem Analysis), PO3 (Design/Development of Solutions), PO4 (Conduct Investigations), PO5 (Engineering Tool Usage), PO6 (Engineer and World), PO7 (Ethics), PO8 (Teamwork), PO9 (Communication), PO10 (Project Management and Finance), and PO11 (Lifelong Learning). PO1 is supported through computational and AI/ML knowledge applied to the SLM prototype.

Outcome	Statement	Indicative Contribution
PO1	Engineering Knowledge	Supporting
PO2	Problem Analysis	High
PO3	Design/Development of Solutions	High
PO4	Conduct Investigations of Complex Problems	High
PO5	Engineering Tool Usage	High
PO6	The Engineer and the World	Moderate
PO7	Ethics	Moderate/High
PO8	Individual and Collaborative Teamwork	High
PO9	Communication	High
PO10	Project Management and Finance	Moderate/High
PO11	Lifelong Learning	High
PSO1	Innovative software/hardware solutions for industry and society	High
PSO2	Modern tools and emerging technologies with ethical and societal consideration	High
PSO3	AI/ML engineering concepts for safe, secure, robust production-ready AI/ML solutions	High; prototype-stage contribution

Knowledge and Attitude Profile: WK2 (data analysis and formal aspects of computing), WK4 (specialist computing/AI knowledge), WK5 (design and operational constraints), WK6 (engineering practice/tools), WK7 (role of engineering in society), WK8 (research-informed critical thinking), and WK9 (ethics, inclusion and professional conduct).

5. Year-Level Learning Outcomes (YLO) and Dimension Alignment

Dimension	Most Relevant YLOs
Critical Thinking (CT)	CT-LO1: Challenge existing knowledge through structured analysis; CT-LO2: Apply systematic observation methods; CT-LO3: Evaluate technical information from multiple sources; CT-LO4: Develop reasoned arguments for engineering decisions and corroborate using AI; CT-LO6: Raise doubts on the status quo of objects and knowledge.
Innovation & Professional Development (IPD)	IPD-LO1: Execute project-based learning assignments independently; IPD-LO3: Understand design thinking and its cultural/social implications; IPD-LO4: Use Generative AI to visualise real-world problems.
Research Experience (RE)	RE-LO1: Conduct literature reviews using AI-powered tools; RE-LO2: Define and justify research problems and develop basic proposals; RE-LO3: Structure mini-projects using appropriate research methods; RE-LO4: Communicate with domain experts/researchers.
Real World Exposure (RWE)	RWE-LO1/2: Interact with society/industry and infer insights; RWE-LO4: Teamwork; RWE-LO7: Cultural aspects of design thinking; RWE-LO8: Professional ethics; RWE-LO9: Awareness of industry practices.
Youth Development (YD)	YD-LO1: Empathy and cultural sensitivity; YD-LO4/5: Civic responsibility, law and social harmony; YD-LO6: Awareness of social implications of engineering solutions using AI.

Primary dimension: Critical Thinking. Secondary dimension: Innovation & Professional Development.

Supporting dimensions: Research Experience, Real World Exposure and Youth Development are evidenced in the learning activities. The primary/secondary labels do not imply that every activity must map to every dimension; mapping should follow the evidence produced.

6. Course Architecture: Four Phases

Phase	Indicative Weeks	Scope	Teaching–Learning Focus	Evidence/Deliverables
Phase 1 – General foundations	Weeks 1–2	Department-agnostic	Design thinking vs. traditional problem solving; desirability–feasibility–viability; five-stage process; biomimicry; observation; initial problem spotting; classroom redesign and CSE-context case discussions.	Reflection sheet, observation notes, first-instinct statement, team case analysis.
Phase 2 – General empathy and problem definition	Week 3	Department-agnostic	5 Whys, root-cause analysis, candidate problem selection, user specificity, problem statement drafting, peer feedback and revision. The activities remain applicable across engineering departments.	5 Whys chain, problem selection matrix, revised problem statement.
Phase 3 – AIML foundations Modules 1–2	Weeks 4–7	CSE-AIML specific	Scout domain/dataset candidates; identify users and stakeholders; emotional journey mapping; structured interviews; qualitative and quantitative research; empathy map/persona; POV and HMW; commit a dataset and design brief. Begin inspecting dataset format, provenance, rights, quality and suitability only after the user problem is articulated.	M1.1–M1.3 and M2.1–M2.3 artefacts; Design Brief: chosen domain, dataset, persona, POV, HMW and initial data audit.
Phase 4 – AIML SLM challenge Modules 3–5	Weeks 8–15	CSE-AIML specific	Ideate and justify SLM design; compare baseline vs domain-specific model choices; prototype; prepare and document dataset; train/fine-tune a compact model; test against a task-specific plan; iterate; analyse risks/limitations; assess technical/social/economic feasibility; prepare a track-aware pitch.	Sheets 1–11, model card/technical note, dataset card, evaluation evidence, demo, pitch and track continuation plan.

7. SLM Project Design Principles

- Problem before model: students must justify the human need and target user before committing to a model architecture.
- Dataset before fine-tuning: record source, license/permission, provenance, domain coverage, language, format, sensitive fields, quality issues and preprocessing decisions.
- Individual ownership: each student keeps an individual engineering log and can explain the data pipeline, model configuration, training/evaluation process and limitations, even when working in a team.
- Right-sized SLM: the objective is a small, domain-appropriate prototype, not a claim to compete with general-purpose frontier models. A compact pretrained model adapted to a curated domain dataset is acceptable; training from scratch is optional where data/compute justify it.
- Evidence over claims: compare with a simple baseline; use train/validation/test separation where applicable; select metrics that fit the intended task; document unsuccessful experiments and limitations.
- Responsible design: do not upload confidential, personal, medical, security-sensitive or licensed data to external services without authorisation. Use anonymisation, access control, citations, consent and disclosure of AI assistance.
- Track readiness: Professional – reproducibility, efficiency and integration; Entrepreneurship – user value, customer segment, viability and pitch; Research – gap, hypothesis, baseline, evaluation, ablation and reproducibility.

8. Pedagogical Approach and Learning Activities

CO	Pedagogical Approach	Learning Activity	Dimension	YLO Mapping
CO1	Interactive exposition, Think–Pair–Share, reflective challenge	Students compare design thinking with solution-first engineering, analyse cases, and identify the five stages.	CT; YD	CT-LO1; CT-LO6; YD-LO1
CO2	Empathy-based fieldwork, structured interviews, peer feedback, social constructivism	Students map users, emotions and workflows; conduct/paraphrase interviews; synthesise qualitative and survey evidence into a persona, POV and HMW.	CT; RE; RWE	CT-LO2; CT-LO3; RE-LO2; RWE-LO1/2
CO3	Problem-based learning, divergent/convergent ideation, mind maps, SCAMPER, Crazy 8s	Students ideate possible dataset/model approaches, compare alternatives, annotate design decisions and explain why a design change responds to user evidence.	CT; IPD	CT-LO4; IPD-LO3/4
CO4	Studio-based prototyping, guided coding, iterative feedback	Students create a data pipeline and compact SLM prototype; record configurations, checkpoints and implementation decisions.	IPD; CT	IPD-LO1; CT-LO4
CO5	Peer review, test-plan design, experiment and evaluation	Students define task metrics, run a baseline comparison, record errors and feedback, revise the prototype and discuss feasibility and risks.	CT; RE; RWE	CT-LO2/3; RE-LO3; RWE-LO8/9
CO6	Pitch practice, business-model workshop, research showcase	Students communicate user value, technical constraints, social/ethical implications and next steps for one chosen track.	IPD; RWE; YD	IPD-LO1/3; RWE-LO4/8; YD-LO6

9. Session-Wise Plan (60 Contact Hours)

Planning basis: 15 weeks × 4 contact hours = 60 hours. The weekly distribution below is an implementation plan that integrates the syllabus units and experiential exercises with the stated AIML SLM outcome. Faculty may adjust actual dates while preserving the sequence, phase boundaries and evidence checkpoints.

Week	Sess.	Topic / Content	Pedagogy / Planned Activity	Phase	CO	Dimension	Evidence / CA Link
Week 1	1	Course orientation; what design thinking is and is not	Interactive lecture; diagnostic reflection; individual starting-point statement	Phase 1 – general	**CO1**	**CT; YD**	Starting-point reflection
Week 1	2	Design thinking principles, origins and human-centred design	Think–Pair–Share; compare solution-first and problem-first cases	Phase 1 – general	**CO1**	**CT**	Comparison notes
Week 1	3	Five stages: Empathise, Define, Ideate, Prototype, Test	Stage-card sequencing; short case walkthrough	Phase 1 – general	**CO1**	**CT**	Stage map
Week 1	4	Awareness challenge: redesign the classroom experience	Individual ideation, pair discussion, quick concept sketch	Phase 1 – general	**CO1, CO3**	**CT; IPD**	Activity 1 artefact
Week 2	1	Desirability, feasibility and viability	Case analysis; stakeholder/value discussion	Phase 1 – general	**CO1, CO3**	**CT; RWE**	Desirability–feasibility–viability notes
Week 2	2	Empathy maps and user observation	Guided observation using department-neutral cases	Phase 1 – general	**CO2**	**CT; RWE; YD**	Observation sheet
Week 2	3	Biomimicry and nature-inspired problem solving	Examples, analogy generation and reflection	Phase 1 – general	**CO1, CO3**	**CT; IPD**	Biomimicry reflection
Week 2	4	IDEO case discussion; problem spotting	Video/case debrief; teams identify who is excluded by a design	Phase 1 – general	**CO1, CO2**	**CT; RWE**	Observation notes and case analysis
Week 3	1	Root cause vs. surface symptom; 5 Whys	Facilitated worked example and individual chain	Phase 2 – general	**CO2**	**CT**	5 Whys chain
Week 3	2	Problem selection matrix: human presence, observability, significance, designability, researchability	Teams score candidate problems and discuss trade-offs	Phase 2 – general	**CO2, CO3**	**CT; RWE**	Selection matrix
Week 3	3	Drafting and testing problem statements	Peer exchange; specificity and root-cause checks	Phase 2 – general	**CO2**	**CT; IPD**	Draft and revised statement
Week 3	4	General-phase gate review and reflection	Faculty review; return to starting-point statement; consolidate problem framing	Phase 2 – general	**CO1, CO2**	**CT; IPD**	Phase 1–2 portfolio checkpoint
Week 4	1	CSE-AIML design lens: digital products, data and language-model use cases	Domain scouting across 2–3 candidate domains	Phase 3 – AIML	**CO2, CO3**	**CT; RWE**	Candidate domain table
Week 4	2	Dataset discovery and domain feasibility	Dataset source, provenance, format, license and access screening	Phase 3 – AIML	**CO2, CO3**	**CT; RE**	Dataset shortlist
Week 4	3	M1.1 awareness challenge and M1.2 empathy warm-up	Complete module sheets; user/data stakeholder mapping	Phase 3 – AIML	**CO1, CO2**	**CT; YD; RWE**	M1.1–M1.2
Week 4	4	M1.3 draft HMW and domain shortlisting	Score curiosity, feasibility and impact; explain shortlist	Phase 3 – AIML	**CO2, CO3**	**CT; IPD**	Draft HMW and shortlist
Week 5	1	Affective design and emotional journey mapping	Map a specific user task with emotion and intensity	Phase 3 – AIML	**CO2**	**RWE; YD**	M2.1 journey map
Week 5	2	Structured interview design and empathetic questioning	Develop neutral questions; discuss consent and respectful research	Phase 3 – AIML	**CO2**	**RE; RWE; YD**	Interview protocol
Week 5	3	Interview practice and synthesis	Conduct approved interview or use a clearly labelled representative-user source; paraphrase notes	Phase 3 – AIML	**CO2**	**RE; RWE**	M2.2 interview log
Week 5	4	Empathy map and persona	Convert observed evidence into needs, constraints, goals and trust concerns	Phase 3 – AIML	**CO2**	**RE; CT**	Persona and empathy map
Week 6	1	Qualitative vs quantitative research; mini-survey design	Draft survey questions, sampling and response plan	Phase 3 – AIML	**CO2**	**RE; CT**	Survey instrument
Week 6	2	Synthesis: themes, evidence and counter-evidence	Triangulate interview/survey findings; avoid unsupported assumptions	Phase 3 – AIML	**CO2, CO3**	**RE; CT**	Theme/evidence matrix
Week 6	3	POV statement, stakeholder map and committed HMW	Workshop to turn evidence into one testable design challenge	Phase 3 – AIML	**CO2, CO3**	**CT; IPD**	POV, stakeholder map, HMW
Week 6	4	Dataset suitability and ethical review gate	Inspect sample rows/documents; record quality, rights, privacy and gaps	Phase 3 – AIML	**CO2, CO5**	**CT; RE; YD**	Dataset audit
Week 7	1	Design Brief review: persona, POV, HMW and dataset	Faculty conference; revise scope and approve domain	Phase 3 – AIML	**CO2, CO3**	**CT; IPD**	Approved Design Brief
Week 7	2	Dataset acquisition, cleaning and train/validation/test planning	Hands-on data preparation with provenance tracking	Phase 3 – AIML	**CO4**	**IPD; CT**	Data pipeline v1
Week 7	3	Task definition: what should the SLM do?	Define input/output examples and task-specific success criteria	Phase 3 – AIML	**CO3, CO5**	**CT; RE**	Task card and examples
Week 7	4	Baseline plan and readiness checkpoint	Choose baseline approach and model route; confirm compute and data access	Phase 3 – AIML	**CO3, CO4**	**CT; IPD**	Phase 3 gate: approved dataset/task
Week 8	1	Unit III: individual user perspective and social experience	Review user narratives; identify multiple stakeholder needs	Phase 4 – AIML SLM	**CO2, CO3**	**RWE; CT**	Problem Spotter Sheet
Week 8	2	Divergent ideation: mind mapping, SCAMPER, Crazy 8s	Generate model, tokenisation, retrieval and workflow alternatives	Phase 4 – AIML SLM	**CO3**	**CT; IPD**	Idea generation sheet

Week	Sess.	Topic / Content	Pedagogy / Planned Activity	Phase	CO	Dimension	Evidence / CA Link
Week 8	3	Convergent ideation and selection criteria	Compare impact, feasibility, risk, data needs and evaluation potential	Phase 4 – AIML SLM	CO3	CT; IPD	Selected concept and rationale
Week 8	4	SLM design options: train-from-scratch vs compact-model adaptation	Explain trade-offs; select a feasible student-level SLM path	Phase 4 – AIML SLM	CO3, CO4	CT; IPD	Model decision record
Week 9	1	Transformer overview and domain-specific design surfaces	Inspect tokenisation, embeddings, attention, positional encoding, objective and size	Phase 4 – AIML SLM	CO3	CT; RE	Architecture analysis
Week 9	2	Architecture hypothesis and annotated diagram	Link each proposed change to a user need and data property	Phase 4 – AIML SLM	CO3, CO4	CT; RE	Annotated architecture sketch
Week 9	3	Prototype V1: data loader, preprocessing and training/evaluation pipeline	Guided implementation; version control and experiment log	Phase 4 – AIML SLM	CO4	IPD; CT	Prototype V1 checkpoint
Week 9	4	Baseline model / baseline task solution	Run a simple baseline and record metrics or task outcomes	Phase 4 – AIML SLM	CO4, CO5	CT; RE	Baseline results
Week 10	1	Compact SLM configuration and resource budget	Choose model size, sequence length, batch size, training method and compute limits	Phase 4 – AIML SLM	CO3, CO4	CT; IPD	Configuration sheet
Week 10	2	Training/fine-tuning lab I	Run a small controlled training/adaptation experiment	Phase 4 – AIML SLM	CO4	IPD; CT	Run log/checkpoint
Week 10	3	Training/fine-tuning lab II and troubleshooting	Inspect loss, overfitting, data leakage and training failures	Phase 4 – AIML SLM	CO4, CO5	CT; IPD	Experiment log
Week 10	4	Evaluation dataset and test plan	Define task metrics, qualitative checks, baseline comparison and failure cases	Phase 4 – AIML SLM	CO5	CT; RE	Test Plan & Feedback Record
Week 11	1	Prototype V1 demonstration and peer/mentor review	Structured feedback against user need and model task	Phase 4 – AIML SLM	CO4, CO5	RWE; IPD	Feedback record
Week 11	2	Failure analysis and model/data revision	Diagnose errors; revise data, prompts, tokenisation or configuration with rationale	Phase 4 – AIML SLM	CO4, CO5	CT; IPD	Revision plan
Week 11	3	Prototype V2 implementation	Implement the highest-priority change and rerun evaluation	Phase 4 – AIML SLM	CO4, CO5	IPD; CT	Prototype V2
Week 11	4	Compare V1 vs V2; record negative results	Analyse metric changes, representative errors and trade-offs	Phase 4 – AIML SLM	CO5	CT; RE	Version comparison
Week 12	1	Responsible SLM design: privacy, bias, safety, licensing and limitations	Risk review and model/dataset documentation	Phase 4 – AIML SLM	CO5, CO6	YD; RWE; CT	Risk register
Week 12	2	Usability/task validation with representative users or structured proxy testing	Observe whether outputs support the intended workflow; do not expose sensitive data	Phase 4 – AIML SLM	CO5	RWE; CT	User/task feedback
Week 12	3	Technical and economic feasibility	Estimate compute, latency, maintenance, data refresh and operating constraints	Phase 4 – AIML SLM	CO5, CO6	IPD; RWE	Feasibility sheet
Week 12	4	Proof-of-concept review	Peer review against SMART success criteria and HMW	Phase 4 – AIML SLM	CO5	CT; RWE	PoC review
Week 13	1	Unit V: target users, value proposition and social benefit	Define users/customers/beneficiaries and what improves for them	Phase 4 – AIML SLM	CO6	RWE; YD	Value proposition
Week 13	2	Business Model Canvas (simplified)	Map users, value, channels, resources, costs, partners and sustainability	Phase 4 – AIML SLM	CO6	IPD; RWE	Canvas
Week 13	3	Track pathways: Professional / Entrepreneurship / Research	Identify next-semester fit, required evidence and gaps	Phase 4 – AIML SLM	CO6	IPD; RE	Track continuation plan
Week 13	4	Project management, milestones and deployment feasibility	Define next steps, risks, ownership and measurable milestones	Phase 4 – AIML SLM	CO6	IPD; CT	Implementation plan
Week 14	1	Pitch storyboard and design showcase preparation	Build a six-panel story: user, friction, insight, concept, evidence, next step	Phase 4 – AIML SLM	CO6	IPD; RWE	Pitch storyboard
Week 14	2	SLM reproducibility and documentation	Package code/configuration, data lineage, model card, dataset card and instructions	Phase 4 – AIML SLM	CO4, CO5	IPD; CT	Reproducibility pack
Week 14	3	Final testing and demo rehearsal	Run final evaluation; verify setup and fallback demonstration	Phase 4 – AIML SLM	CO5, CO6	CT; IPD	Final test record
Week 14	4	Peer review and final iteration	Make bounded final corrections; record remaining limitations	Phase 4 – AIML SLM	CO5	CT; RWE	Final prototype
Week 15	1	Final Design Showcase: problem and user evidence	Present persona, POV, HMW and dataset justification	Phase 4 – AIML SLM	CO2, CO6	CT; RWE	Showcase part 1
Week 15	2	Final Design Showcase: SLM demo and evaluation	Demonstrate individual SLM artefact and explain baseline/metrics/errors	Phase 4 – AIML SLM	CO4, CO5	IPD; CT	SLM demo
Week 15	3	Final pitch: feasibility, responsible use and chosen track	Defend value, limitations and track continuation pathway	Phase 4 – AIML SLM	CO6	IPD; RWE; YD	Pitch and Q&A
Week 15	4	Portfolio audit, reflection and course closure	Submit complete portfolio; revisit initial statement and reflect on changed mindset	Phase 4 – AIML SLM	CO1–CO6	CT; IPD; YD	Final portfolio and reflection

10. Assessment Strategy Design

The syllabus specifies a 100% Continuous Assessment model: Unit-wise Exercise Portfolio (80%) and Comprehensive Design Challenge Project (20%). The portfolio captures the student's process, including reflection and problem discovery; the comprehensive project integrates the design brief, dataset, SLM prototype, evaluation and pitch. Since the end product is a technical artefact, the assessment must reward the traceability from user evidence to model decisions, not only model performance.

Component	Weight	Mode	Assessment Evidence	COs
A. Unit-wise Exercise Portfolio	80%	Continuous, staged submission	Phase 1–2 general activities; Phase 3 M1.1–M2.3 foundations; Phase 4 sheets and checkpoints. Evidence includes reflections, observations, 5 Whys, selection matrix, persona, interview/survey synthesis, POV/HMW, dataset audit, idea selection, architecture hypothesis, prototypes, test results, revision log, risk register and reflection.	CO1–CO5
B. Comprehensive Design Challenge Project	20%	Final integrated project + demonstration/pitch	Individual SLM artefact with dataset/model documentation, baseline and task-specific evaluation, user/problem traceability, iteration evidence, limitations, responsible-use review, feasibility and track-aware pitch.	CO2–CO6
TOTAL	100%			CO1–CO6

11. Recommended Rubrics

Criterion	Marks (%)	Evidence / Performance Basis
Process portfolio completeness and reflection	10	Sequence, completeness, revision evidence, reflective insight
Empathy and research quality	15	Specific user, ethical data collection, quality of questions, evidence-based synthesis
Problem definition and HMW/POV	10	Root cause, user specificity, observable need, actionable scope
Dataset choice and data governance	10	Domain relevance, provenance/license, quality, privacy and preprocessing rationale
Ideation and architecture rationale	10	Divergent/convergent thinking, alternatives, traceability to user/data evidence
Prototype and individual SLM implementation	15	Runnable or reproducibly demonstrated artefact, individual contribution, documented configuration
Testing, evaluation and iteration	15	Baseline, appropriate metrics, representative failure cases, version comparison and honest limitations
Responsible design and feasibility	5	Risk/ethics analysis, resource and deployment constraints
Communication and track pathway	10	Clear demo/pitch, value proposition, limitations, actionable next-semester plan
TOTAL	100	

The recommended rubric distributes the 80% portfolio and 20% project into evidence-rich criteria; faculty should map the criteria to the official gradebook weights without double-counting. A practical implementation is to score the portfolio criteria across the semester and use the final integrated project criteria for the 20% capstone.

12. Assessment Calendar and Gate Checks

Window	Focus	Required Evidence / Gate
Weeks 1–3	General foundations	Portfolio checkpoint 1: starting point, empathy/observation, 5 Whys, problem selection and revised problem statement.
Weeks 4–6	AIML Modules 1–2: empathy and research	Portfolio checkpoint 2: domain shortlist, user persona, interview/survey synthesis, POV/HMW.
Week 7	Dataset/task approval	Gate 1: approve domain, dataset source/rights, task definition, user need and feasible SLM scope.
Weeks 8–9	Ideation and design	Checkpoint 3: selected concept, annotated architecture hypothesis, dataset audit and baseline plan.
Weeks 10–11	Prototype and iteration	Checkpoint 4: SLM prototype V1/V2, run logs, feedback and comparison evidence.
Weeks 12–13	Validation and feasibility	Checkpoint 5: evaluation, risk register, feasibility, value proposition and track continuation plan.
Weeks 14–15	Final project	Final portfolio, individual SLM demonstration, technical note/model card, pitch and reflection.

13. AI Integration and Academic Integrity

Policy Level	Guidance
Level 1 – Prohibited for the activity	Personal reflections, observed user responses and individual demonstration should not be fabricated or delegated to AI. Do not claim interviews that did not occur.
Level 2 – Permitted with disclosure	AI may assist preliminary literature/dataset discovery, question refinement, code explanation, debugging suggestions and documentation drafting where permitted. Record tool, purpose and verification.
Level 3 – Encouraged with disclosure	AI may be used for brainstorming alternatives, code assistance, data exploration, evaluation scripts and visualisation, provided the student verifies outputs and documents use.
Mandatory disclosure	Include a short AI-use statement listing tools, tasks supported, generated code/text retained, verification performed, and known errors. Students must explain the artefact independently.
Data protection	Do not upload identifiable interview notes, confidential datasets, restricted domain data or credentials to public AI tools. Use synthetic/de-identified data or approved institutional systems.

14. Required Student Submissions

- Design Thinking portfolio covering all assigned general and worksheets, with dates and version history.
- One-page Design Brief: domain, intended user, task/workflow, persona, evidence-backed POV, HMW and committed dataset.
- Dataset card: source, license/permission, provenance, format, size, language/domain, preprocessing, sensitive-data checks and known limitations.
- Model/SLM technical note: base model or architecture, why it is suitable, adaptation/training method, configuration, compute, checkpoints and reproducibility instructions.
- Evaluation package: baseline, split strategy, metrics, representative success/failure examples, revision comparison and limitations.
- Annotated architecture or pipeline diagram, prototype V1/V2 evidence, test plan, peer/user feedback and risk register.
- Final demonstration/pitch and track continuation plan (Professional, Entrepreneurship or Research).
- Individual contribution statement and AI-use disclosure.

15. Course Outcome to Assessment Mapping

CO	Assessment Evidence	Coverage	Rationale
CO1	Portfolio: Phase 1–2 reflections and foundations worksheets	Primary	Design thinking principles and stages
CO2	Portfolio: empathy, interviews, survey, persona, POV/HMW	Primary	User research and problem definition
CO3	Portfolio: ideation, model/dataset alternatives, design rationale	Primary	Systematic idea generation and convergence
CO4	Portfolio + comprehensive project: dataset pipeline and SLM prototype	Primary	Proof-of-concept construction
CO5	Portfolio + comprehensive project: test plan, metrics, iteration, feasibility	Primary	Validation and evaluation

CO	Assessment Evidence	Coverage	Rationale
CO6	Comprehensive project: pitch, value proposition and track plan	Primary	Entrepreneurship/social innovation and progression

16. Track Readiness: Semester-to-Semester Continuity

Track	Evidence Carried Forward
Professional Track	A working, reproducible domain SLM prototype; clean data pipeline; latency/compute estimates; evaluation report; integration/ API or deployment sketch; reliability and maintenance risks.
Entrepreneurship Track	Validated user pain point; target segment and value proposition; prototype demo; basic Business Model Canvas; operating-cost assumptions; differentiation; social impact and responsible-use case.
Research Track	Literature matrix; research gap; hypothesis; baseline; task-specific metrics; ablation/controlled comparison where feasible; error analysis; reproducibility pack and next research question.

The semester deliverable is a validated small-scale prototype, not a claim of production readiness. The next semester's chosen track should determine the extension: hardening/integration for Professional, validation of a sustainable value proposition for Entrepreneurship, or deeper experimental investigation for Research.

17. References and Learning Resources

- Brown, T. (2019). Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation. Harper Business.
- Kumar, V. (2012). 101 Design Methods: A Structured Approach for Driving Innovation in Your Organization. Wiley.
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Course Facilitator